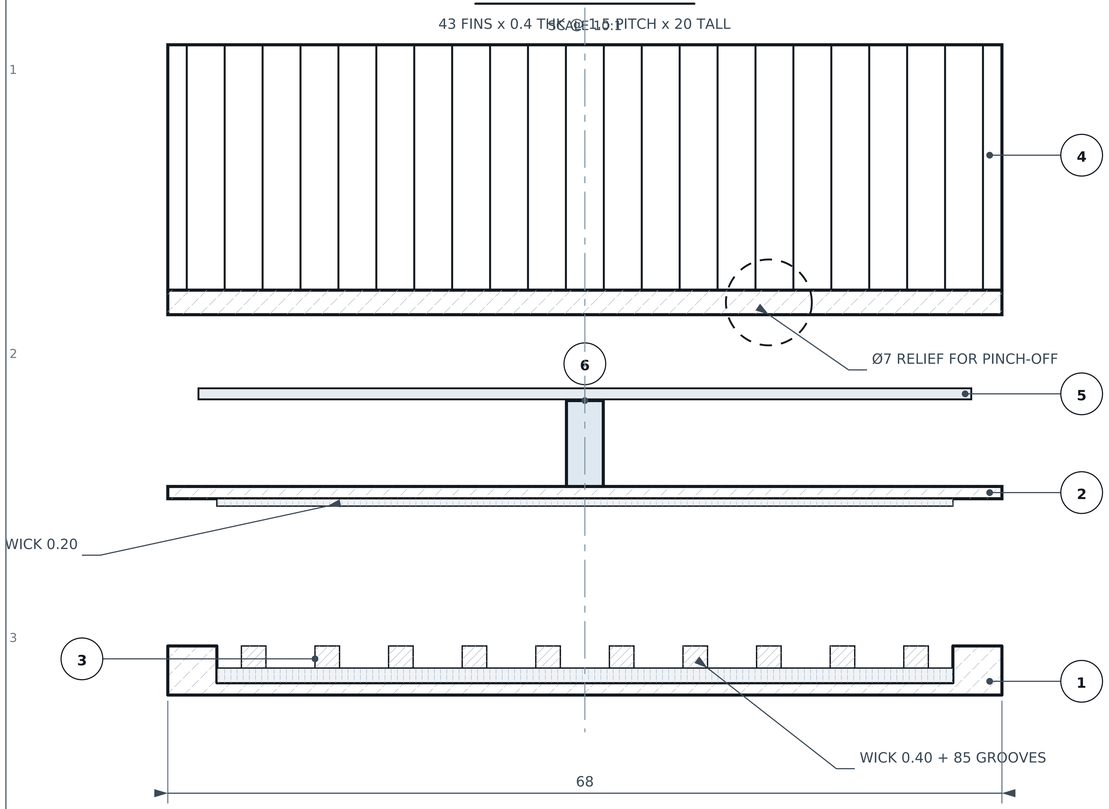
EXPLODED ASSEMBLY

SCALE 10:1



WICK THICKNESS SHOWN x3 FOR CLARITY / ASSEMBLED HEIGHT 27.0

THERMAL RESISTANCE NETWORK

JUNCTION TO AIR, 20 x 20 DIE

DIE	TIM	Cu FLOOR	EVAP	VAPOUR	COND	LID+SPRD	TIM	FIN→AIR	AIR
—	0.025	0.006	0.100	0.004	0.012	0.036	0.011	0.470	25 C

SERIES TOTAL R th(j-a) = 0.67 K/W @ 2 m/s → Q = 113 W at Tj 100 C / air 25 C

Vapour chamber contributes 0.194 K/W (29%); the fin stack dominates. Evaporator wick is the largest single internal term — do not thin the floor wick below 0.40.

BILL OF MATERIALS — ASSEMBLY VC-100

ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	MASS
1	1	VC-101	CHAMBER BASE, POCKETED, 100 POSTS	Cu C10200 / C1020	77.4 g
2	1	VC-102	CHAMBER LID, 1.0 THK, FILL PORT	Cu C10200 / C1020	41.4 g
3	1	VC-103	SINTERED WICK — IN-SITU PROCESS	Cu POWDER 45-75um	9.6 g
4	1	VC-201	FIN STACK, 43 FIN — FORCED AIR	Al 1100-H14	88.1 g
4A	1	VC-202	FIN STACK, 13 FIN — NAT. CONVECTION	Al 1100-H14	72.7 g
5	1	VC-204	THERMAL INTERFACE PAD 0.2 THK	PCM / SIL-PAD	0.4 g
6	1	VC-205	PINCH-OFF TUBE OD3.0 x ID2.0 x 25	Cu C10200 / C1020	1.2 g
7	1.90 mL	VC-206	WORKING FLUID — DEAERATED	DI H2O 18 MOhm.cm	1.87 g

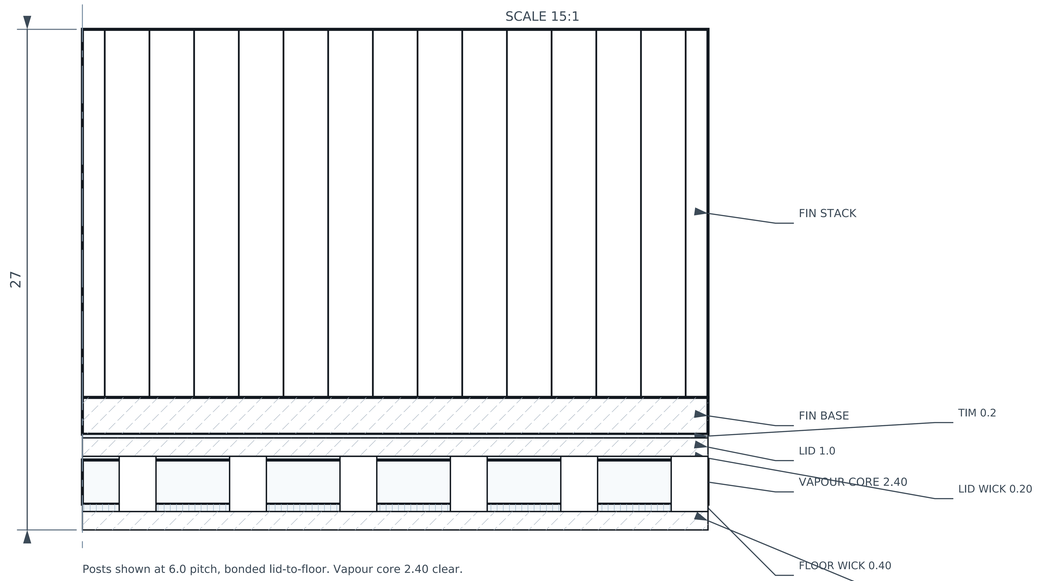
TOTAL MASS, CHARGED: 218 g (CONFIG A) / 203 g (CONFIG B)

PERFORMANCE — 20 x 20 DIE, Tj 100 C, AIR IN 25 C

CONFIGURATION	AIR	R th (j-a)	Q @ dT 75 K
CONFIG A — 43 fin @ 1.5 pitch	1 m/s	1.12 K/W	**67 W**
CONFIG A — 43 fin @ 1.5 pitch	2 m/s	0.67 K/W	**113 W**
CONFIG A — 43 fin @ 1.5 pitch	4 m/s	0.46 K/W	**163 W**
CONFIG B — 13 fin @ 5.0 pitch	natural	3.42 K/W	22 W
CONFIG B — passive, Tj 85 C limit	**natural**	**—**	**16.5 W**
capillary limit — water, vertical	—	—	804 W
capillary limit — water, horizontal	—	—	858 W

SECTION X—X (HALF, FROM CENTRELINE)

SCALE 15:1



Posts shown at 6.0 pitch, bonded lid-to-floor. Vapour core 2.40 clear.

GENERAL NOTES

1. Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
2. Charging, evacuation and sealing per VC-300. NEVER charge before diffusion bonding — bond temperature exceeds the fluid critical pressure envelope.
3. CONFIG A requires forced air >= 1 m/s. For still air use CONFIG B (VC-202); a 1.5 mm fin pitch chokes natural convection to h = 0.03 W/m2K.
4. Orientation-independent: 10.5 kPa capillary head vs 0.66 kPa gravity head over 68 mm.

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **GENERAL ASSEMBLY** — VAPOUR CHAMBER + FIN STACK / Cu-H2O / 68 x 68 x 27	SCALE: **10:1**	UNITS: **mm**	THIRD ANGLE PROJECTION
DRAWING No.: **VC-100**	REV: **A**	SHEET: **1 / 7**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-09